



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

PROJECT PROGRESS 1

Subject: Programming For Bioinformatics (SECB3202)

Session: 2025/2026 Semester 1

Title: Pancreatic Cancer Prediction

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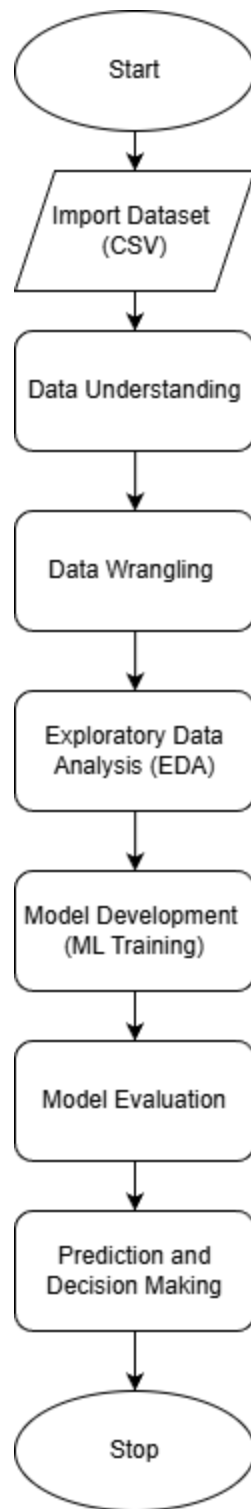
1.0 Software Requirement

Software	Version	Purpose
Python	Python 3.12	Main programming language for the project
Pandas	Latest	Data manipulation and preprocessing
NumPy	Latest	Numerical operations
Matplotlib	Latest	Data visualization
Scikit - Learn	Latest	Machine learning model development and evaluation
Jupyter Notebook	Latest	Coding environment, testing and documentation
GitHub	N/A	Version control and project progress tracking
VS Code	Latest	Python IDE

2.0 Hardware Requirement

Hardware	Minimum Requirement	Recommended
Laptop / PC	i3 / Ryzen 3	i5 / Ryzen 5
RAM	4 GB	8 GB and above
Storage	1 GB Free Space	1 GB Free Space or Higher
GPU	Not required	NVIDIA GPU (<i>optional for faster ML</i>)
Internet	Dataset download purpose	Stable Internet

3.0 Flowchart



Flowchart 1.0: Flow of the Project

The flowchart begins with importing the pancreatic cancer dataset in CSV format. Once the data is loaded, the next step is to understand the dataset by examining the variables, data types, and overall structure. After gaining an initial understanding, data wrangling is performed to handle missing values, encode categorical variables, and clean the dataset for further analysis.

Exploratory Data Analysis (EDA) is then conducted to observe patterns, correlations, and distributions within the data. Insights from EDA help guide the machine learning phase, where classification models are developed and trained. After the models are built, their performance is evaluated using appropriate metrics. The results from these evaluations form the basis of prediction and decision-making, concluding the overall workflow.